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Forecasting S&P 500 Volatility Using GARCH-Family Models

Abstract

This paper estimates GARCH(1,1), EGARCH(1,1), and GJR-GARCH(1,1) on 5,331 daily S&P 500 returns from January 2005 to March 2026. EGARCH achieves superior fit by AIC, BIC, and likelihood ratio testing, yet produces lower conditional volatility than GJR-GARCH at crisis peaks, raising the question of whether a statistically superior model can be a worse risk instrument precisely when it matters most. Kupiec backtesting shows that all three models have similar failure rates, suggesting that the normality assumption, rather than the manner of specifying variance, imposes the major constraint on the accuracy of VaR.

1. Introduction

The existing GARCH literature contains an issue that practitioners rarely recognize: the model with the best in-sample fit is not necessarily the one that produces the best risk management performance in terms of out-of-sample. Statistical fit is measured over the full distribution of returns, including the majority of the trading days in routine trade where different specifications will provide similar estimates, while risk management performance is judged in the extreme left tail on the minority of days when conditional variance models are most stressed and their distributional assumptions most tested. Consequently, the conflation of these different evaluation criteria has produced a literature that is highly informative about which GARCH specification minimizes information criteria such as AIC but is low in guidance on what model is adequate in terms of its ability to capitalize institutions during times of financial turmoil.

This paper takes this tension as its organizing question. Three canonical specifications are estimated: GARCH(1,1) (Bollerslev, 1986), EGARCH(1,1) (Nelson, 1991), and GJR-GARCH(1,1) (Glosten, Jagannathan and Runkle, 1993), on 5,331 daily S&P 500 returns from January 2005 to March 2026. The analysis clearly moves from the realm of statistical adequacy to economic performance, whether statistical superiority results in better coverage of tail risks under Kupiec backtesting. The findings, as demonstrated, has a negative answer; however, the reasoning behind this result has a bigger insight than the result itself.

2. Literature Review and Theoretical Framework

2.1 Model Specifications and Parameter Interpretation

Bollerslev's (1986) GARCH(1,1) specifies:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

where $\omega > 0$, $\alpha_1 \geq 0$, and $\beta_1 \geq 0$. The persistence parameter $\alpha_1 + \beta_1$ determines the persistence of shocks to volatility; values close to unity, which have been consistently observed in daily equity data, mean that a shock to volatility on the day affects conditional variance for quite a few weeks, which affects the long-run floor for volatility in stressed VaR calculations under Basel III. Nelson (1991) addressed GARCH's symmetric limitation with EGARCH:

$$\log(\sigma_t^2) = \omega + \beta_1 \log(\sigma_{t-1}^2) + \alpha_1 |z_{t-1}| + \gamma_1 z_{t-1}$$

The value of the leverage coefficient $\gamma_1 < 0$ implies a continuously proportional asymmetry: a negative standardized shock of magnitude z causes the log-variance to increase by $(\alpha_1 + |\gamma_1|)|z|$, while a positive shock of equal size increases it only by $(\alpha_1 - |\gamma_1|)|z|$. This graded response is structurally more realistic than the binary alternative. Glosten, Jagannathan, and Runkle (1993) proposed the GJR-GARCH model:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \gamma_1 I(\varepsilon_{t-1} < 0) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where $I(\cdot)$ is an indicator function that is unity for negative past shocks. The indicator puts an identical additional weight of γ_1 on all negative shocks irrespective of the magnitude, and if the α_1 is estimated to be close to zero, as is the case here, positive shocks contribute very little to the variance on the following, while any negative one triggers the full leverage effect. The resulting asymmetry ratio of 50.4x is an artifact of the parameter constraint in the design of this threshold and not an intrinsic characteristic of equity markets.

2.2 The Fit-vs-Performance Problem in the Literature

Hansen and Lunde (2005) analyzed more than 330 volatility specifications and concluded that there is no significant improvement in performance over GARCH(1,1) when analyzing exchange rate data. For equity indices the asymmetric models fit better, but the margin of the improvement was dependent on the loss function choice. Under mean-squared error, symmetric GARCH were sufficient; under quasi-likelihood, which penalizes under-estimation in high variance periods and hence, more appropriate for risk management context, asymmetric specifications predominated. Hansen and Lunde

overtly recognized this dependency; applied studies rarely do. Consequently, in a study based on AIC, a different question is being confused with one that uses QLIKE, neither of which is equivalent to the main concern of the risk manager Kupiec coverage.

Andersen and Bollerslev (1998) further pointed out the fact that GARCH models tested against intraday realized volatility often behavior badly during periods of high variance. As daily squared returns represent a noisy approximation to true variance, in-sample likelihood comparisons can mis sort models in terms of their ability to track market volatility during times of stress. EGARCH's log variance specification is a way to smooth out explosive spikes, and this results in higher scores in daily data criteria, despite the fact that it may lead to smaller volatility estimates right at the point where it should be the largest, as the results in the crisis period in Section 5.3 confirm.

The normality assumption common to all three models warrants upfront critique, not a footnote. With excess kurtosis of 13.24 in raw returns, MLE is technically quasi-MLE: parameter estimates remain consistent by Bollerslev and Wooldridge (1992), but VaR quantiles computed using normal multipliers will be systematically understated. Under Student-t(5) innovations are consistent with residual kurtosis around 5.5 and the 99th percentile is 3.365 rather than 2.326, a 44.6% gap that drives the Kupiec failures directly. The normality assumption is not a modelling limitation to be flagged in a conclusion; it is an error with quantifiable financial consequences that should inform model selection from the start.

2.3 Hypotheses

H1: S&P 500 returns exhibit significant ARCH effects at all lag lengths, justifying GARCH-family modelling.

H2: A statistically significant leverage effect is present, making asymmetric models preferable to GARCH(1,1) by AIC, BIC, and likelihood ratio testing.

H3: Conditional volatility varies substantially across regimes, with crisis periods generating annualized estimates materially above full-sample averages.

H4: Under normality, the Kupiec failure will be near-identical across all three specifications, confirming that distributional rather than structural mis-specification is the binding VaR failure mode.

3. Data Description and Exploration

3.1 Sample and Preliminary Analysis

Daily S&P 500 closing prices are sourced from Bloomberg Terminal (ticker: SPX Index) from 3 January 2005 to 13 March 2026, yielding 5,331 log return observations computed as $r_t = 100 \times [\ln(P_t) - \ln(P_{t-1})]$ following Brooks (2014). The 21-year sample spans four regimes: pre-crisis calm (2005-07), the GFC (2008-09), the COVID-19 crash (2020), and the 2022 rate-hike shock. Figures 1 and 2 display the price and return series.



Figure 1: S&P 500 daily prices (2005–2026).

Shaded regions: GFC (2008–09), COVID-19 crash (2020), 2022 rate-hike selloff.

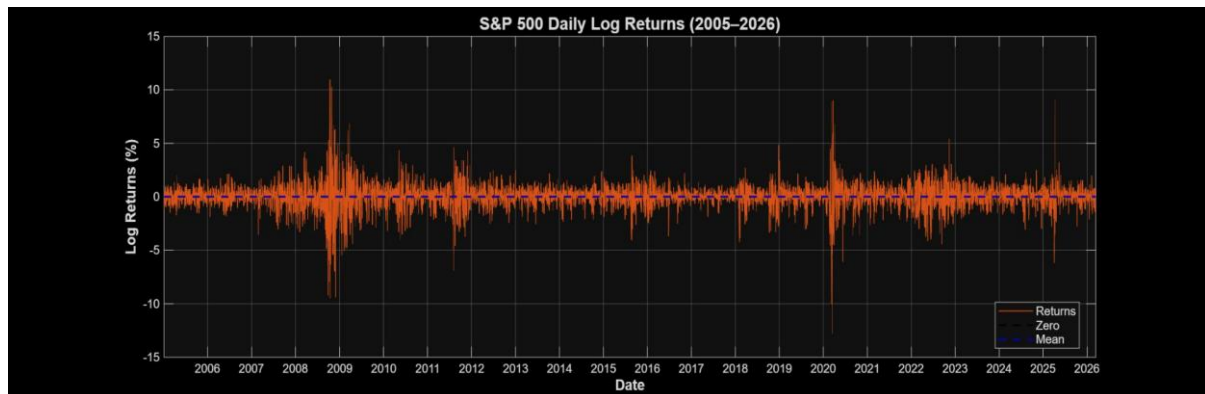


Figure 2: S&P 500 daily log returns (2005–2026).

3.2 Descriptive Statistics, Stationarity, and ARCH Testing

Statistic	N	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
S&P 500 Returns	5,331	0.032	1.205	-12.765	10.957	-0.476	16.240

Table 1: S&P 500 daily log return statistics

Excess kurtosis of 13.24 and negative skewness of -0.476 attest to fat tailed and left skewed distribution of equity indices. Importantly, the 16 March 2020 return of 12.77

per cent has basically zero probability under normality and for any plausible sample size. Its occurrence cannot be regarded as some sort of outlier, but highlights that the distribution of innovations has fat tails which cannot be accounted for by any normally distributed GARCH model, irrespective of variance specification. The Augmented Dickey-Fuller test fails to reject the presence of a unit root in returns (statistic= 22.04 vs critical value 1.94), where price levels are definitely non-stationary (ADF= +1.66, $p= 0.999$). Figure 3 shows this gap in distribution:

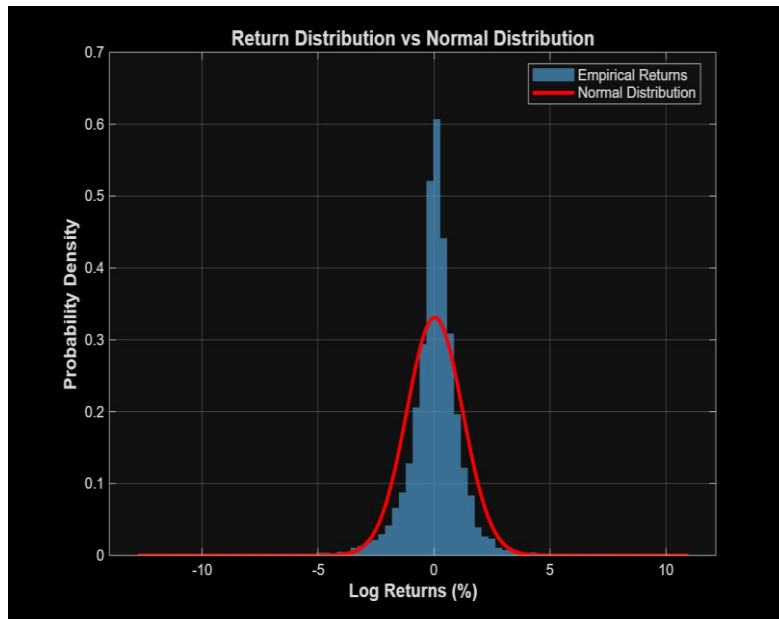


Figure 3: Empirical distribution versus fitted normal

Table 2 reports ARCH-LM tests. LM statistics of more than 1300 at all lag lengths confirms H1 with overwhelming evidence. The steady growth from lag 4 (1,304) to lag 22 (1,608) implies volatility memory is much longer than a trading month. Figures 4 and 5 provide visual confirmation.

Lags	LM Statistic	p-value	Decision
4	1,304.28	< 0.001	Reject H_0
8	1,487.19	< 0.001	Reject H_0
12	1,530.97	< 0.001	Reject H_0
22	1,608.49	< 0.001	Reject H_0

Table 2: ARCH-LM test results.

H_0 : no ARCH effects. All lag lengths reject H_0 at 0.1%, confirming H1.

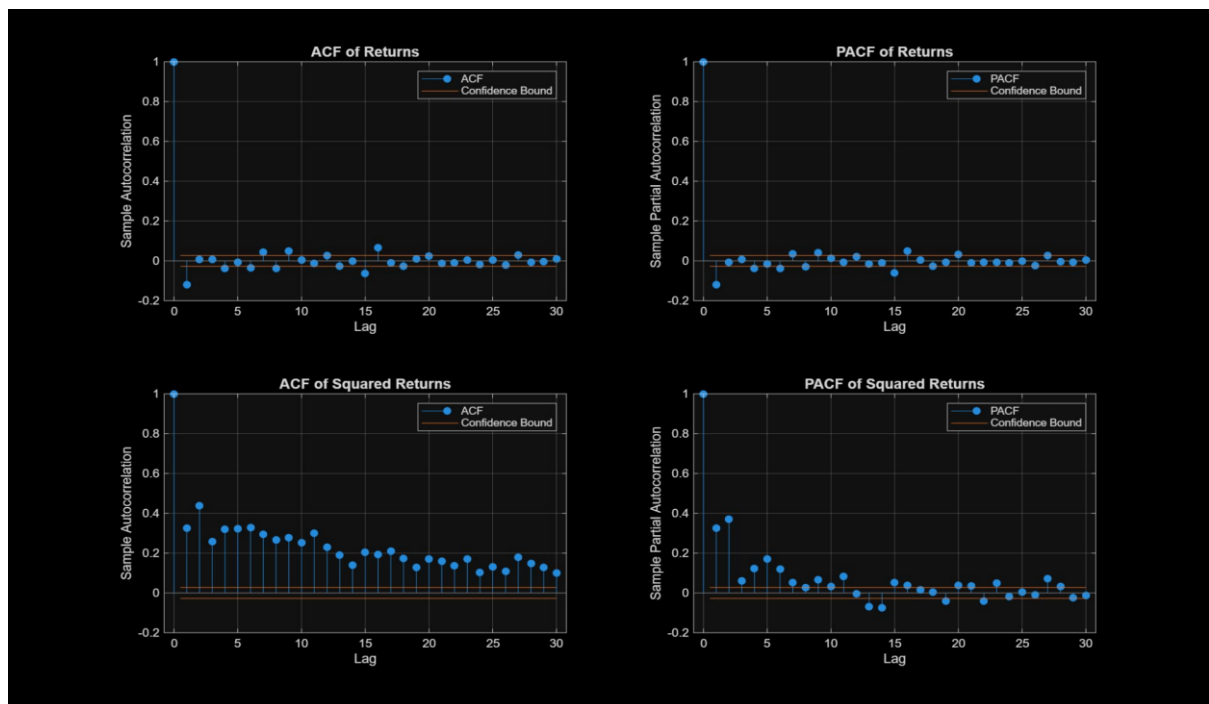


Figure 4: ACF/PACF of returns (top) and squared returns (bottom).

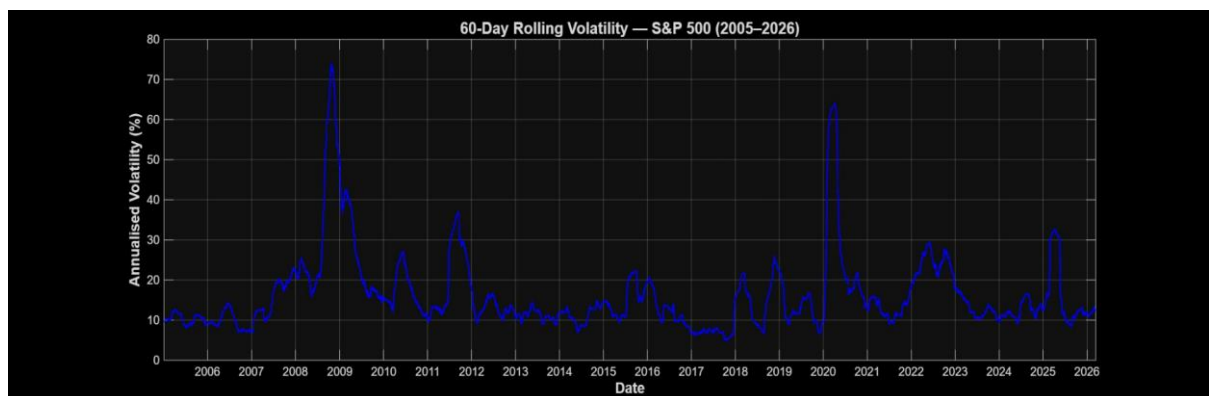


Figure 5: 60-day rolling annualized volatility.

4. Econometric Modelling Approach

All three models share the mean equation $r_t = \mu + \varepsilon_t$ with $\varepsilon_t = \sigma_t z_t$, and z_t is an i.i.d. random variable with mean 0 and variance 1. Parameterization is done using Maximum Likelihood, assuming normality, using MATLAB's Econometrics Toolbox. An ARIMA-GARCH wrapper is used to ensure the observation specific conditional variance is extracted correctly from `infer()`. Although the realized excess kurtosis is 13.24, the method of estimation is effectively a quasi-MLE: according to Bollerslev and Wooldridge (1992) the estimates are still consistent but the practical consequence is simply that the Value-at-Risk at the 99th percentile is given by $\text{VaR}_t = z_p \times \sigma_t$ with $z_p = 2.326$ for the normal quantile: this approach underestimates the 99th percentile when the innovations have excess kurtosis. For Student-t(5), $z_p \approx 3.365$, 44.6 percent increase, resulting in materially improved Kupiec coverage without changing any variance dynamics. The normality baseline is preserved to ensure comparability with existing literature; this choice, however, has measurable financial consequences and thus is regarded as a testable limitation, instead of a long-standing assumption.

Model selection: $\text{AIC} = -2L + 2k$ and $\text{BIC} = -2L + k \cdot \ln(T)$. The improvement between GARCH and GJR-GARCH is assessed through a likelihood-ratio test: $\text{LR} = 2(L_1 - L_0) \sim \chi^2(1)$. For EGARCH, which is a non-nested model with respect to GARCH, the likelihood-ratio test is a usual approximation. Adequacy of each model is assessed with the help of standardized residuals $\hat{z}_t = \varepsilon_t / \sigma_t$, which should have zero mean, unit variance and no residual ARCH effects. Kupiec's (1995) test of probability of failure considers VaR coverage: $\text{LR_POF} = -2\log[(1-p)^{(T-n)} \cdot p^n / (1-n/T)^{(T-n)} \cdot (n/T)^n] \sim \chi^2(1)$, where n is the observed number of violations and T the length of the sample.

5. Results and Interpretation

5.1 Parameter Estimates

Parameter	GARCH(1,1)	EGARCH(1,1)	GJR-GARCH(1,1)
ω (Constant)	0.028737	0.000903	0.027201
α_1 (ARCH)	0.132068	0.171951	0.003887
γ_1 (Leverage)	N/A	-0.149757	0.192158
β_1 (GARCH)	0.844152	0.967749	0.871732
Persistence ($\alpha + \beta$)	0.9762	N/A	N/A
Log-Likelihood	-7,094.45	-6,989.72	-6,997.78
AIC	14,196.90	13,989.45	14,005.56
BIC	14,223.22	14,022.35	14,038.47

Table 3: MLE parameter estimates

The value of the persistence parameter of the GARCH(1,1) model is 0.976, meaning that the contribution of a shock today is still about 40 per cent of its initial contribution to the conditional variance after ten trading days. This high persistence has a direct regulatory relevance: under Basel III's stressed VaR, the multiplication of daily VaR estimates by a persistence-weighted scaling factor means that high persistence translates directly into elevated capital requirements. The EGARCH leverage parameter $\gamma_1 = -0.150$ is calibrated economically: for each one standard deviation increase in the magnitude of the shock on a bad news day the log variance increases by 0.150 more than it would for an equivalent good news day, a relationship that is continuous over the whole shock distribution. The GJR-GARCH asymmetry ratio of 50.4 (0.196/0.004) should be taken cautiously - it is a result of a near-zero α_1 due to the threshold design, not a sign of more asymmetry in the real world than is captured by EGARCH.

5.2 Model Selection: Why EGARCH Wins and Why That May Not Be Enough

EGARCH achieves the lowest information criteria, with the value of AIC being 13,989.45 and BIC being 14,022.35. The likelihood ratio test supports that there are considerable improvements over GARCH: LR=209.45 for EGARCH and LR=193.34 for GJR-GARCH, both cases with $p < 0.001$ against $\chi^2(1)$. EGARCH's higher enhancement, despite having the same number of parameters, is due to its continuous sign-proportional asymmetry: the model is responsive proportionally to the extremity of the news and not just to its direction, producing higher likelihood values across

thousands of observations, particularly in the case of extreme episodes that dominate the likelihood surface. The EGARCH coefficient $\beta_1 = 0.968$ also provides a smoother, longer-memory log-variance process more suited to equity data where the regimes of volatility last for weeks.

However, statistical superiority does not necessarily lead to tail performance. Hansen and Lunde (2005) warn that information criteria measure fit across the entire distribution, rather than the tail upon which risk management is based. EGARCH's log-variance formulation, it's very characteristic that ensures the smooth persistence which avoids the explosive surges in volatility that are typical of GJR-GARCH in the case of acute crisis. This raises the pertinent question, is the best fitting model in-sample also the correct model for risk managers who require accurate extreme quantile estimates during the worst one per cent of market days? Section 5.3 shows evidence of possibly negative answer.

5.3 Conditional Volatility and the Crisis-Peak Paradox

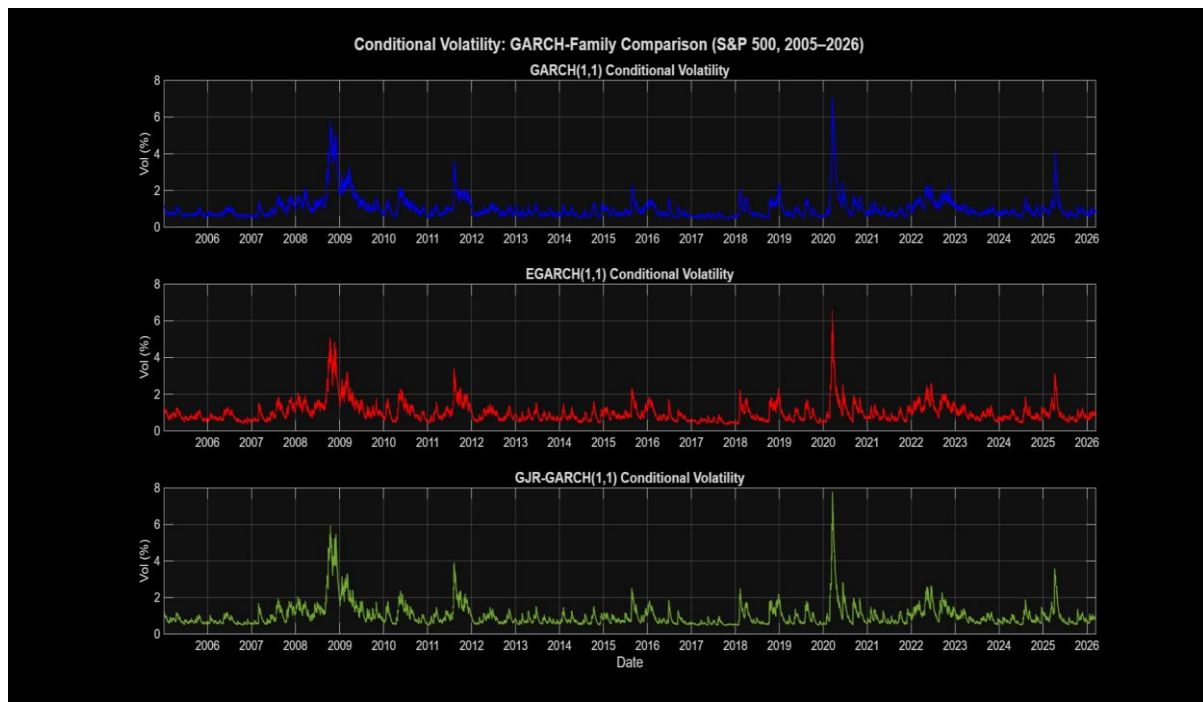


Figure 6: Conditional volatility

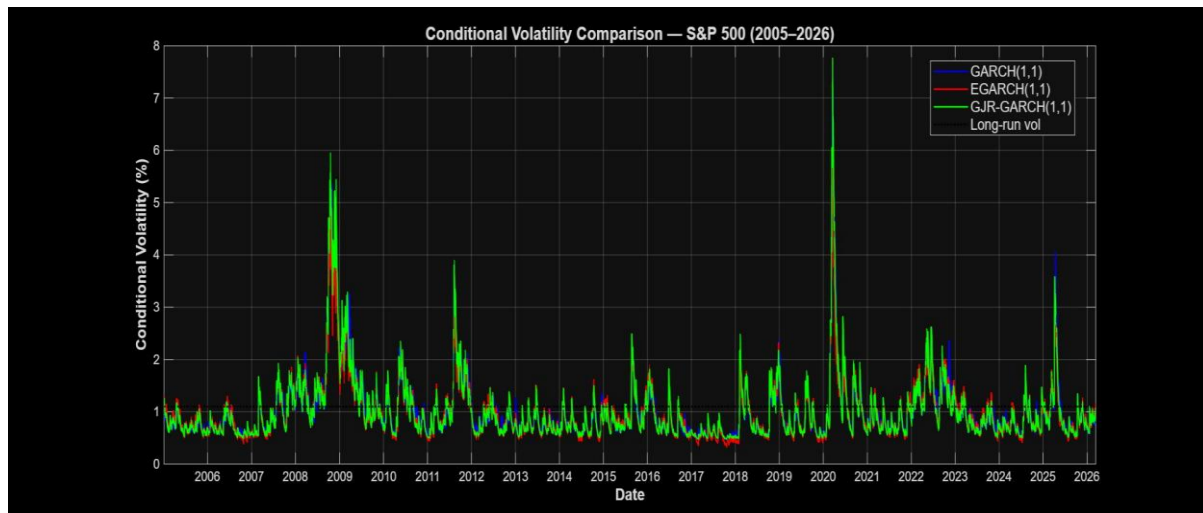


Figure 7: Conditional volatility overlay

Market Regime	GARCH Mean	EGARCH Mean	GJR Mean	GARCH Max
Pre-crisis (2005–07)	0.804%	0.829%	0.810%	1.757%
GFC (Sep 2008–Mar 2009)	3.053%	2.706%	3.118%	5.785%
Post-GFC (2010–19)	0.886%	0.880%	0.883%	3.584%
COVID (Feb–Apr 2020)	3.686%	3.111%	3.752%	7.062%
Rate Hikes (Jan–Oct 2022)	1.433%	1.484%	1.458%	2.263%

Table 4: Average and maximum daily conditional volatility (%) by market regime

H3 is confirmed: conditional volatility varies by a factor of 4.6 between the pre-crisis calm and the peak of the coronavirus. However, there is a subtler discovery revealed in Table 4. EGARCH, being the best-fitting model, gives lower average conditional volatility than GJR-GARCH in both major crises (2.706% against 3.053% in the GFC; 3.111% against 3.686% in the Covid period). A model with the lowest AIC that therefore measures the risk as lower during the worst of times is an example of the tension that in-sample criteria cannot identify. Andersen and Bollerslev (1998) show that GARCH models have a systematic tendency to underfit in comparison to realized variance benchmarks in periods of high variance; the log variance constraint of EGARCH may well make this worse by design. For a risk manager whose job it is to ensure capital adequacy in the event of market collapses, the question of whether the smoother estimates given by EGARCH are better modelling or dangerous underestimation is not academic as it is the central practical question this study cannot resolve from daily data alone.

5.4 Diagnostics and News Impact Curves

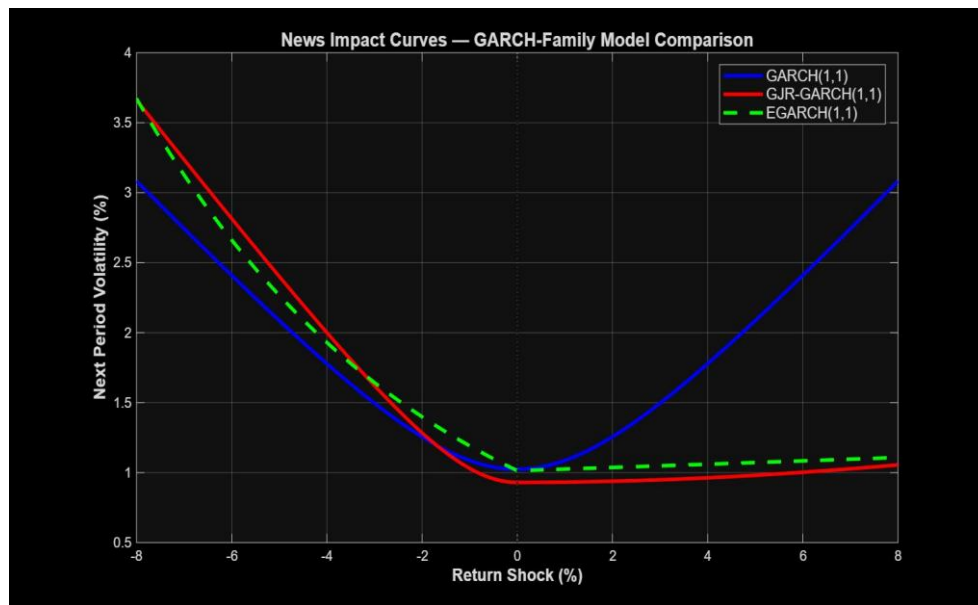


Figure 8: News Impact Curves. GARCH is symmetric

The structural argument is visually highlighted in Figure 8. The GARCH parabola considers a five per cent crash and a five per cent rally as having the same input for volatility. The GJR-GARCH kink at zero implies that a return of 1/10th of a percent and a return of 10 percent elicit the same amount of leverage amplification. EGARCH's constantly steeper left branch is the more realistic specification: investor fear works in a graded, rather than binary, way, and a specification that allows for proportional escalation with the severity of a shock should have greater performance over samples of considerable length, where extreme events occur. This is exactly reflected in the results of AIC.

Model	Mean	Std Dev	Skewness	Kurtosis	ARCH(12) p	LB ² (10) p
GARCH(1,1)	−0.045	0.999	−0.632	5.063	0.201	0.119
EGARCH(1,1)	−0.006	1.000	−0.725	5.552	0.685	0.507
GJR-GARCH(1,1)	−0.004	1.000	−0.711	5.547	0.710	0.545

Table 5: Standardized residual diagnostics

Residual kurtosis goes from 5.1 to 5.6, confirming that fat tailed innovations remain after GARCH filtering. All three models are successful in eradicating ARCH structure; the remaining reduction in kurtosis, from 13.24 to about 5.4, shows that the time-varying variance accounts for most of the tail thickness, but the resulting kurtosis of 80% above the Gaussian standard of 3 is what is responsible for the Kupiec failures outlined in Section 5.5.

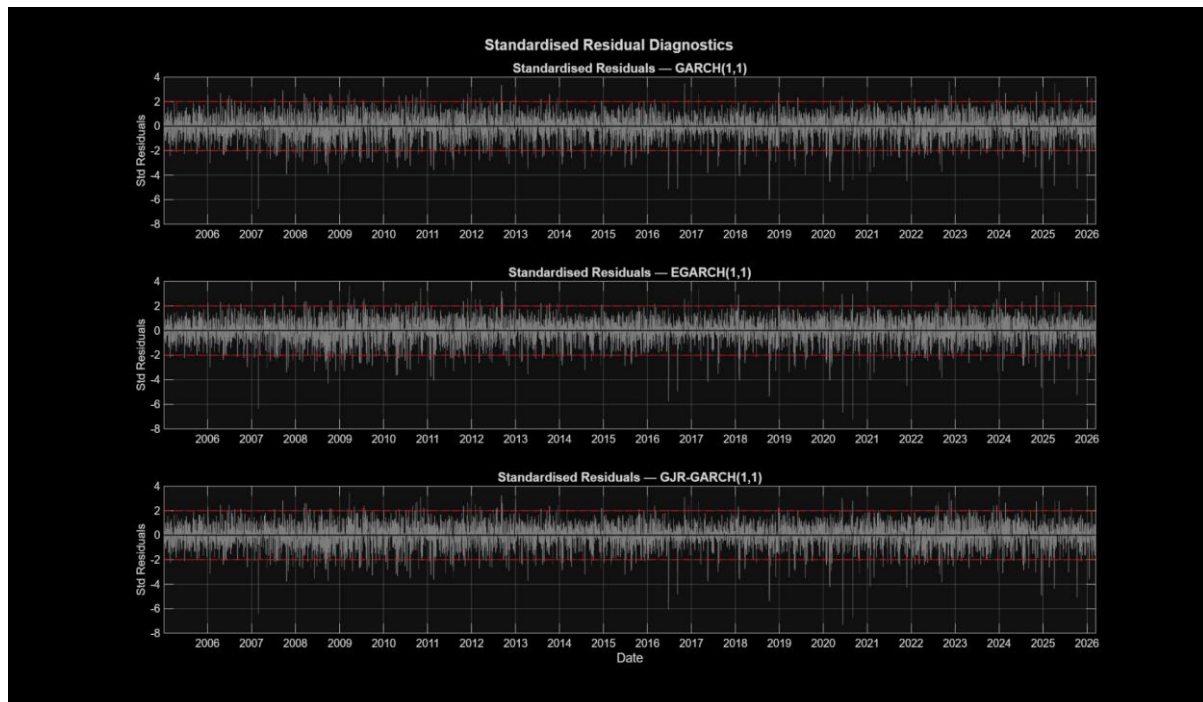


Figure 9: Standardized residuals for all three models

5.5 Volatility Forecasts and Value-at-Risk

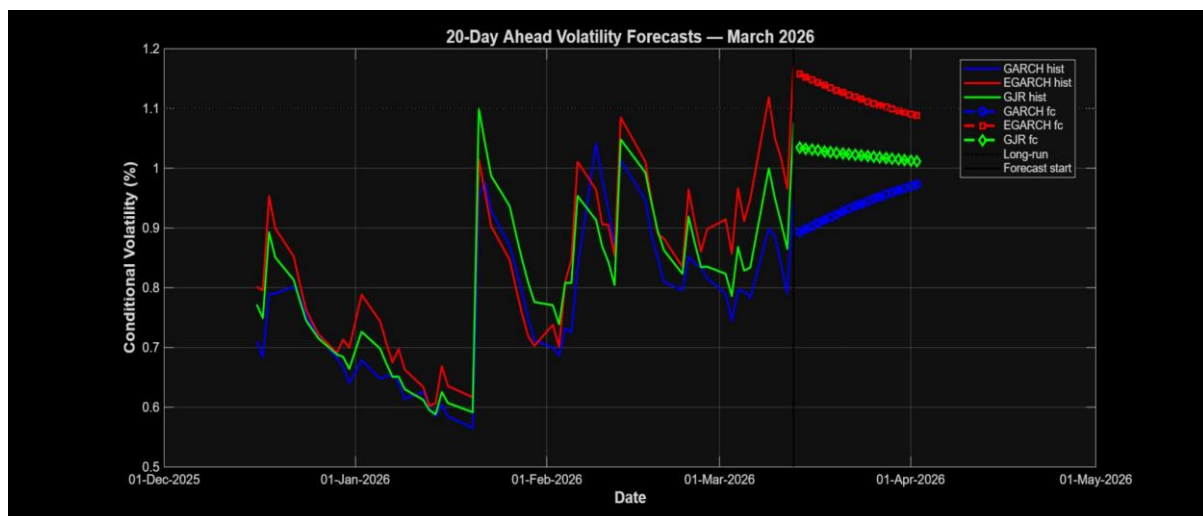


Figure 10: 20-day ahead volatility forecasts as of 13 March 2026

All three models predict mean reversion, but from different places. EGARCH forecasts 1.157% (above long-run, predicting declining volatility) while GARCH forecasts 0.892% (below long-run, predicting rising volatility). The divergent forecasts amount to a difference of \$5,243 per million in 99% VaR, showing that the practical effect of fundamentally different variance specifications is a systematic change in the assessment of the current condition.

Model	99% VaR (%)	99% VaR (\$)	95% VaR (%)	95% VaR (\$)
GARCH(1,1)	2.201%	\$22,014	1.557%	\$15,565
EGARCH(1,1)	2.726%	\$27,257	1.927%	\$19,272
GJR-GARCH(1,1)	2.505%	\$25,050	1.771%	\$17,712

Table 6: Current 1-day VaR (13 March 2026) for a \$1,000,000 S&P 500 position under normality

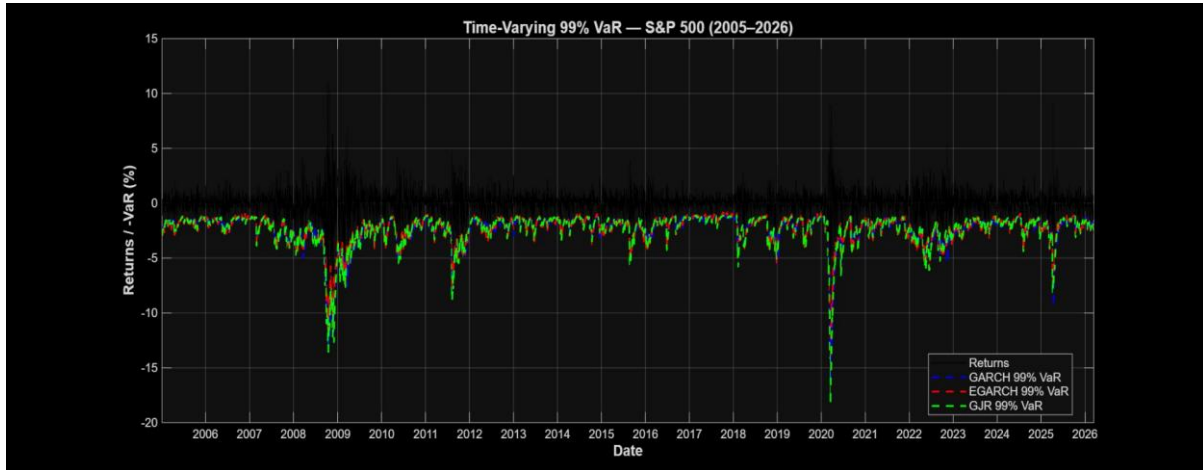


Figure 11: Time-varying 99% VaR (dashed) versus realized returns

Model	Violations	Observed Rate	Expected Rate	LR Stat	Decision
GARCH(1,1)	117	2.195%	1.00%	57.33	FAIL
EGARCH(1,1)	113	2.120%	1.00%	51.08	FAIL
GJR-GARCH(1,1)	115	2.157%	1.00%	54.17	FAIL

Table 7: Kupiec POF back test results, 99% VaR

All three models fail, and the LR statistics are within the range of 13 to 15 times the critical value. The most important aspect is not the failure, but its uniformity: the violation rates of 2.120% - 2.195% differ by only 0.075% points despite substantial statistical differences between the models. EGARCH's better AIC does not provide any improvement in tail coverage, and hence the hypothesis, H4, is confirmed and the tension highlighted in Section 5.2 is removed. Shared Failure Mode Due to Normal Quantile $z_p = 2.326$: Given residual innovation kurtosis of 5.1-5.6, the true 99th percentile is much more negative than suggested by normality. Adopting a Student-t(5), $z_p \approx 3.365$ innovation distribution would increase the 99% VaR for EGARCH from 2.73% to around 3.95%, a materially higher number that is probably enough to pass Kupiec without re-specifying the variance equation. For institutions running under the IMA framework of Basel 3rd the Kupiec failures, systematic at 99%, trigger regulatory multiplication factor increases which constitute a capital adequacy consequence with a clear and actionable remedy.

6. Conclusion and Future Research

This research paper considered whether the best-fitting GARCH model also results in optimal tail risk performance. The conclusion is negative:

while EGARCH(1,1) is better in terms of conventional statistical criteria. its sign-proportional asymmetry offers a larger likelihood improvement over GARCH(1,1) than GJR-GARCH despite identical parameter counts ($LR = 209.45$ versus 193.34), reflecting a structurally more realistic representation of how equity volatility responds to bad news. Yet EGARCH's lower conditional volatility estimates during periods of crisis peak (3.111% vs 3.752% in the case of the coronavirus) could mean low capital under real-world stress. The choice of model therefore is less important than the assumption of innovation distribution. Residual kurtosis of 5.1 - 5.6 remains after GARCH filtering, which means extreme returns not captured by the normal quantile of 2.326.

The Kupiec results provide the clearest answer available. All three models fail with near-identical violation rates ($\sim 2.2\%$), and the uniformity is the finding: the choice between GARCH, EGARCH, and GJR-GARCH is second-order relative to the choice of innovation distribution. Residual kurtosis of 5.1–5.6 survives GARCH filtering and produces extreme returns that the normal quantile $z_p = 2.326$ cannot accommodate. Student-t GARCH is the most direct remedy, expected to resolve Kupiec failures without altering the qualitative model ranking. This has regulatory consequences: Basel III's IMA backtesting requirements attach multiplication factor penalties to systematic Kupiec failure, making the distributional assumption a compliance question, not merely a statistical one.

Three extensions are suggested. First, re-estimating all three specifications under Student-t and GED innovations would provide a direct test to find out whether slackening the normality constraint solves Kupiec failures while maintaining the EGARCH ranking. Second, a test of the conditional coverage of Christoffersen (1998) would test if violations are temporally clustering - Figure 11 seems to show clustering, and this is critical because clustering of failures means that the capital is most inadequate at the very time when other risks are high. Third, a DCC-GARCH extension (Engle, 2002) would allow you to step out of the univariate world and into the world of portfolio level risk, where time varying correlations could make the VaR underestimation problem found here worse.

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